

A Ground-Based Telemetry Analysis Pipeline for CubeSat Anomaly Detection with Synthetic Fault Injection

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Abstract—Reliable anomaly detection in spacecraft telemetry is essential for mission assurance, yet labelled fault datasets from real hardware tests remain scarce. This paper presents a modular ground-based telemetry analysis pipeline developed for the KISPE Satellite Learning Laboratory (SLL), a CubeSat-class hardware-in-the-loop platform. The pipeline automates parsing of SCOTTI archive files, cleans and decodes binary telemetry fields, applies four complementary anomaly detectors — linear thermal trend analysis, z-score flagging, cross-channel correlation breakdown, and stuck-sensor detection — and generates up to twenty diagnostic plots per experiment session. Applied to nine completed subsystem test campaigns (totalling 3,340 CDH and 3,392 ADCS telemetry packets), the pipeline consistently identifies a universal on-board data handling (OBDH) thermal ramp of $0.8^{\circ}\text{C min}^{-1}$ to $2.0^{\circ}\text{C min}^{-1}$ across all sessions, 21 thermally-stuck sensor channels, and experiment-specific voltage-rail anomalies. To support future machine-learning classifier development, a synthetic fault injection framework generates 16 labelled anomaly types — spanning thermal, power, ADCS, reaction wheel, and timing fault classes — producing 71 labelled CSV artefacts per experiment from real baseline telemetry. The full pipeline, injector, and dataset are released openly alongside this paper.

Index Terms—CubeSat, anomaly detection, telemetry analysis, fault injection, satellite health monitoring, ADCS, synthetic dataset generation

I. INTRODUCTION

Spacecraft anomaly detection has traditionally relied on threshold monitors and expert-defined rules embedded in ground software. While effective for known, well-characterised faults, such approaches struggle to generalise to novel failure modes and require extensive manual configuration for each new mission. The emergence of machine-learning (ML) anomaly detection — in particular unsupervised and semi-supervised methods — offers a path toward more adaptive, data-driven fault identification [?], [?]. A fundamental bottleneck, however, is the availability of labelled fault datasets from real satellite hardware; most operational anomaly datasets are proprietary, sparsely labelled, or gathered only in simulation.

The KISPE Satellite Learning Laboratory (SLL) addresses this gap from the hardware perspective. It is a laboratory-scale, CubeSat-derived platform that provides a fully integrated Command and Data Handling (CDH) and Attitude Determination and Control System (ADCS) in a representative flight-like configuration, with real-time telemetry streamed via

the SCOTTI ground interface. In this work we exploit the SLL to conduct nine distinct subsystem test campaigns and to develop a complete software pipeline for:

- 1) **Telemetry parsing** — ingesting both Analysis-format and Debug-format SCOTTI archives, with automatic fallback;
- 2) **Data cleaning** — decoding signed/unsigned hex fields, deduplicating packets, deriving elapsed time, and flagging stuck sensor channels;
- 3) **Anomaly detection** — applying four complementary statistical detectors that operate on CDH and ADCS streams;
- 4) **Visualisation** — generating up to 20 diagnostic plots per session for exploratory analysis;
- 5) **Synthetic fault injection** — producing a labelled, ML-ready dataset by injecting 16 fault types into the cleaned real-world baseline telemetry.

The contributions of this paper are:

- A reproducible, open-source ground-test telemetry pipeline for CubeSat-class hardware;
- A characterisation of naturally occurring anomalies observed across nine SLL test campaigns;
- A systematic fault injection taxonomy and implementation covering thermal, power, ADCS, wheel, and timing fault classes;
- A labelled dataset of 71 CSV artefacts per experiment, derived from real hardware telemetry, suitable for training and evaluating anomaly detection models.

The remainder of the paper is structured as follows. Section ?? reviews related work. Section ?? describes the SLL platform and telemetry format. Section ?? details the pipeline architecture. Section ?? describes the test campaigns. Section ?? presents detection results. Section ?? describes the synthetic fault injection framework. Section ?? discusses findings and limitations, and Section ?? concludes.

II. BACKGROUND AND RELATED WORK

A. Spacecraft Anomaly Detection

Early spacecraft health monitoring relied on hard-coded limit checks applied to individual telemetry channels [?]. Subsequent work introduced correlation-based monitors, which flag deviations in pairs of channels that are normally tightly

coupled [?]. Statistical process control methods, including Shewhart charts and exponentially weighted moving averages, have been applied to satellite voltage and temperature streams [?].

B. Machine Learning Approaches

Long Short-Term Memory (LSTM) networks have shown particular promise for multivariate spacecraft telemetry anomaly detection, with the SMAP and MSL Mars rover datasets becoming de-facto benchmarks [?]. Telemanom and related prediction-error methods flag time steps where a trained sequence model cannot reconstruct the observed signal. More recently, transformer-based architectures [?] have been applied to satellite telemetry, though they demand larger labelled corpora than are typically available from single missions.

C. Synthetic Data and Fault Injection

The lack of labelled spacecraft fault data has motivated several synthetic generation strategies. [?] benchmarked statistical and deep methods on simulated Mars rover data. Fault injection at the software level — inserting error patterns directly into clean sensor streams — has been used in avionics and industrial process control to create balanced training sets [?]. To the authors’ knowledge, no prior work has applied a systematic, multi-class fault injection framework to real CubeSat hardware telemetry at the subsystem test-campaign level.

III. PLATFORM AND TELEMETRY FORMAT

A. KISPE Satellite Learning Laboratory

The KISPE SLL is a CubeSat-derived educational and research platform comprising an On-Board Computer (OBC), an Electrical Power System (EPS) with battery and solar simulation, a Communications module (COMMS), and a full ADCS board hosting a 3-axis accelerometer, gyroscope, magnetometer, sun sensors, a reaction wheel, and magnetorquers. The platform generates two telemetry streams at approximately 1 Hz–2 Hz:

CDH packets contain power rail voltages and currents (SYS_V, PLAT_5V_V, OBC_PWR_V, COMMS_PWR_V, ...), nine board temperatures (TEMP_SOLAR through TEMP_PAYLOAD), 20 expansion thermistor channels, and housekeeping counters.

ADCS sensors provide inertial measurements (ACCEL, GYRO, MAG), eight sun sensor channels, sun angle, wheel speed, and magnetorquer commands.

B. SCOTTI Archive Format

The SCOTTI ground interface writes telemetry to two file types per session:

Analysis files contain full decoded fields in human-readable key-value format (), one packet per timestamped block, with sub-millisecond timestamps.

Debug files record raw hex payloads together with space-delimited “Plot” lines that carry all 87 CDH and 30 ADCS

decoded values in fixed positional order. Debug files use date formatting and 1 s timestamp precision, compared to the sub-millisecond format of Analysis files. Experiments where the Analysis file was empty (Magnetorquer_1, AttitudeDeterminationMagnetometer_1, AttitudeDetermination_Control, CameraTest_1) were ingested via the Debug fallback.

IV. PIPELINE ARCHITECTURE

The pipeline is implemented in Python 3 using pandas, NumPy, SciPy, and Matplotlib, and is structured into five modules: , , , and . Fig. ?? illustrates the data flow.

A. Parsing ()

The parser applies a regular-expression state machine to scan sequential lines of an archive file. For Analysis files, a timestamp-header regex identifies each packet boundary, and a tab-indented field regex extracts name-value pairs. For Debug files, a single line regex matches Plot lines and maps the space-delimited tokens to 87 CDH or 30 ADCS field names using fixed positional schemas. Hex strings that cannot be converted to float are preserved as strings for downstream handling. The parser returns two objects (CDH and ADCS), sorted by timestamp.

B. Cleaning ()

Cleaning applies several transformations:

- 1) **Hex decoding:** unsigned-integer fields (current readings) are converted via ; signed 16-bit fields (BATT_CHR_I, WHEEL_SPEED, MAGNETORQUER_*) use two’s-complement decoding. Mixed string/float columns produced by duplicate Debug sessions are handled by checking the input type before attempting hex parsing.
- 2) **Deduplication:** rows with identical timestamps () are removed. This eliminates the duplicate sessions present in several Debug files.
- 3) **Elapsed time:** a scalar column is derived from the packet timestamp minus the session start.
- 4) **Power estimation:** estimated power in milliwatts is computed for four supply rails (SYS, PLAT_5V, OBC_PWR, COMMS_PWR) from voltage $\times$ decoded current.
- 5) **Stuck thermistor detection:** any thermistor channel with zero variance across all packets is labelled stuck.

C. Detection ()

Four detectors are applied to each cleaned session:

D1 — Linear thermal trend. A least-squares line is fitted to TEMP_OBDH and TEMP_COMMS vs. elapsed time using . The slope ($^{\circ}\text{C}/\text{min}$), R^2 , and total temperature rise ΔT are reported.

D2 — Z-score flagging. For each of seven CDH channels (TEMP_OBDH, TEMP_COMMS, TEMP_SOLAR, SYS_V, PLAT_5V_V, OBC_PWR_V, COMMS_PWR_V), samples with $|z| > 2.5\sigma$ are flagged, where z is computed over all clean samples in the session.

D3 — Correlation breakdown. A rolling Pearson correlation ($w = 30$ samples) is computed for four

TABLE I: SLL Test Campaign Overview

Experiment	Source	CDH	ADCS	ADCS Live
Baseline_1	Analysis	333	335	No
Accelerometer_1	Analysis	360	363	Yes
Gyro_1	Analysis	267	264	Yes
ReactionWheel_1	Analysis	357	359	Yes
SunSensorTest_1	Analysis	361	361	Yes
Magnetorquer_1	Debug	535	534	Yes
AttitudeDet. Magnetometer_1	Debug	285	285	Yes
AttitudeDetermination_Control	Debug	273	273	Yes
CameraTest_1	Debug	545	546	Yes
Total		3,316	3,280	

board-temperature pairs that are physically expected to track closely: (TEMP_EPS, TEMP_BATTERY), (TEMP_EPS, TEMP_BACKPLANE), (TEMP_COMMS, TEMP_PAYLOAD), and (TEMP_ADCS, TEMP_WHEEL). Windows where $|r| < 0.5$ are flagged as correlation breakdowns.

D4 — Stuck-channel detection. Any ADCS sensor channel with zero variance but a non-zero stuck value (excluding the all-zero baseline state) is identified as a frozen sensor.

D. Plotting ()

Up to 20 plots are generated per session. Plots 01–15 target CDH data (board temperatures, OBDH ramp, power rails, thermistor anomaly, correlation matrices, voltage overlays, etc.). Plots 16–20 are conditional on live ADCS data and cover IMU 3-axis time-series, vector magnitudes, reaction wheel speed, sun sensor readings, and magnetorquer commands.

V. EXPERIMENTAL CAMPAIGNS

Nine test campaigns were conducted on the KISPE SLL between 02 March and 22 March 2026, each lasting between approximately 3 min and 10 min. Table ?? summarises the dataset.

The Baseline_1 session was conducted with ADCS powered off as a control experiment. Subsequent campaigns powered the full ADCS stack and exercised individual actuators and sensors in isolation. The CameraTest_1 and AttitudeDetermination sessions were the longest (approximately 9 min) and were captured only in Debug format due to a logging configuration change on 22 March 2026.

VI. RESULTS

A. Universal OBDH Thermal Ramp

The most prominent system-wide finding is a consistent linear rise in the OBDH board temperature across every experiment. Table ?? reports the fitted slope, goodness-of-fit R^2 , and total temperature delta for all nine sessions.

All sessions exhibit $R^2 > 0.88$, confirming highly linear thermal behaviour. The Gyro_1 session shows the steepest rate ($2.04^\circ\text{C min}^{-1}$), while Magnetorquer_1 has the lowest slope but highest total ΔT (8.98°C) owing to its longer duration. CameraTest_1 achieves a ΔT of 10.66°C — the

TABLE II: OBDH Thermal Ramp Characteristics per Session

Experiment	Slope ($^\circ\text{C min}^{-1}$)	R^2	ΔT ($^\circ\text{C}$)
Baseline_1	1.52	0.975	4.48
Accelerometer_1	1.92	0.945	6.09
Gyro_1	2.04	0.948	4.85
ReactionWheel_1	1.56	0.943	5.24
SunSensorTest_1	1.26	0.944	4.24
Magnetorquer_1	0.80	0.888	8.98
AttitudeDet. Magnetometer_1	1.25	0.921	6.97
AttitudeDetermination_Control	1.18	0.933	6.06
CameraTest_1	1.06	0.956	10.66
Mean $\pm$ SD	1.40 ± 0.39	0.939 ± 0.024	6.51 ± 2.06

TABLE III: Z-Score Anomaly Counts ($|z| > 2.5$) for Key Channels

Experiment	SYS_V	PLAT_5V	OBC_PWR	COMMS
Baseline_1	3	3	9	0
Accelerometer_1	17	19	17	2
Gyro_1	18	24	3	19
ReactionWheel_1	7	20	14	23
SunSensorTest_1	6	3	2	4
Magnetorquer_1	16	9	17	2
AttitudeDet. Magnetometer_1	21	13	5	9
AttitudeDetermination_Control	12	0	3	0
CameraTest_1	3	0	0	0

largest observed — consistent with the payload being powered throughout the 9 min session.

This ramp is not an injected anomaly; it reflects steady-state power dissipation in the OBC in the absence of active thermal control, and serves as a reliable discriminant between operational and anomalous thermal states.

B. Stuck Thermistor Array

All experiments register 21 stuck thermistor channels: 16 rod thermistors frozen at 77.18°C and four bath/expansion thermistors frozen at 74.31°C , with the heater thermistor reading zero. These values are characteristic of an unconnected or open-circuit ADC input at the conversion resolution of this firmware version, and represent a systemic sensor interface fault rather than a transient event. CameraTest_1 is the sole exception, with only one stuck channel, suggesting a firmware or cabling update between the March 12 and March 22 campaigns.

C. Voltage Rail Z-Score Anomalies

Table ?? summarises the z-score anomaly counts for the four primary voltage rails. SYS_V and OBC_PWR_V show elevated flag counts in most sessions, reflecting sub-cycle transients on the battery-backed supply rail — a known characteristic of the SLL’s charge-controller interaction at low solar input. PLAT_5V_V anomalies are highest in the Gyro_1 and ReactionWheel_1 sessions, consistent with increased current demand from the ADCS peripherals.

D. ADCS Sensor State

The Baseline_1 session shows all nine IMU channels (AC-CEL_X/Y/Z, GYRO_X/Y/Z, MAG_X/Y/Z) identically zero,

confirming that the ADCS board was unpowered. All subsequent sessions show live ADCS data. Two frozen-sensor findings are notable:

- **SUN_ANGLE frozen at 0x013C (316)** — detected in the Accelerometer_1 and Magnetorquer_1 sessions. The sun angle derivation algorithm was likely not re-initialised between the two sessions, causing the output register to hold the last computed value.
- **WHEEL_SPEED frozen near zero** — observed in Gyro_1. The reaction wheel was not commanded during this session and the speed register did not update, indicating a firmware reporting oversight rather than a mechanical fault.

E. Temperature Correlation Breakdown

The rolling-correlation detector flags the (TEMP_EPS, TEMP_BACKPLANE), (TEMP_COMMS, TEMP_PAYLOAD), and (TEMP_ADCS, TEMP_WHEEL) pairs in all nine sessions. This pattern is consistent across experiments, suggesting that these pairs are not, in fact, tightly correlated on the SLL time-scale of a few minutes — the boards heat at slightly different rates due to their varying power dissipation. This is a useful baseline characterisation: a true thermal decoupling anomaly (e.g. a heater failure) would need to be distinguished from this normal intra-session decorrelation.

F. Packet Timing

Sessions parsed from Analysis files exhibit a mean CDH inter-packet interval of ≈ 0.50 s with a standard deviation of ≈ 0.14 s, reflecting the 2 Hz nominal telemetry cadence. Debug-format sessions show exactly 1.00 s intervals with zero standard deviation, an artefact of the Debug format’s integer-second timestamp resolution. No pathological packet gaps exceeding 2 s were observed.

VII. SYNTHETIC FAULT INJECTION

A. Motivation and Design Principles

Real spacecraft fault datasets are intrinsically imbalanced: nominal operation dominates, and most fault types have very few or zero observed instances. To address this for ML classifier training, we develop a fault injection framework operating directly on the cleaned real-world telemetry. This preserves the realistic noise characteristics, sensor correlations, and thermal drift of the actual hardware while adding controlled, labelled perturbations.

Each injection function is a pure function of its input, returning a modified copy alongside a label record containing the anomaly type, start/end row indices, severity (*low/medium/high*), and affected channel names. All random decisions use a seeded, making every injection fully reproducible.

B. Injection Taxonomy

Table ?? lists the 16 implemented injection types across five fault classes.

TABLE IV: Synthetic Fault Injection Taxonomy

ID	Name	Description
<i>Thermal (CDH)</i>		
T1	OBDH ramp accel.	Accelerate OBDH slope by $\times 4$ from a mid-session split
T2	Board step change	Step δT on a board temp with exponential decay ($\tau = 30$ samples)
T3	Board divergence	Grow a linear bias between two normally-correlated boards
<i>Power (CDH)</i>		
P1	SYS_V sag	Gaussian-shaped battery sag; co-dips BATT_CHR_V
P2	OBC voltage spikes	8 random-sign spikes in a 20-sample OBC_PWR_V window
P3	Rail stuck high	Force ADCS_5V_V to 3.3 V from a random mid-session row
<i>ADCS (ADCS stream)</i>		
A1	Accel axis dropout	Zero ACCEL axes for a contiguous window
A2	Gyro saturation	Peg a gyro axis at its observed maximum
A3	Magnetometer flip	Invert all MAG axes for a short burst
A4	IMU all-zero dropout	Zero all 9 IMU channels for 25% of session
A5	Sun sensor saturation	Set all SUN_SENSOR channels to 1023
<i>Wheel (ADCS stream)</i>		
W1	Wheel overspeed	Linearly ramp WHEEL_SPEED to ± 1800 counts
W2	Wheel sudden stop	Zero WHEEL_SPEED one sample after peak
<i>Timing (CDH)</i>		
C1	Packet gap	Remove rows totalling 10 s from mid-session
<i>Compound</i>		
CP1	Thermal + power fault	T1 + P1 applied simultaneously
CP2	ADCS + power fault	A4 on ADCS + P3 on CDH simultaneously

C. Dataset Statistics

For each of the nine processed experiments, two variants of each injection type are generated using different random seeds, yielding $16 \times 2 = 32$ injected data files and 32 corresponding label CSVs per experiment. Additionally, two clean evaluation copies (with labels) and one are written, bringing the total to 71 artefacts per experiment and $71 \times 9 = 639$ artefacts across the full campaign dataset. Each label record encodes the anomaly type, affected channels, severity level, and exact row span, providing rich supervision signal for both binary and multi-class fault classifiers.

VIII. DISCUSSION

A. Key Findings

The universal OBDH thermal ramp (mean $1.40^\circ\text{C min}^{-1}$, $R^2 > 0.88$ in all sessions) is the dominant feature of SLL telemetry and can serve as a reliable normality baseline: any session that does not exhibit this trend would be immediately anomalous. The distinct rates across experiments ($0.80^\circ\text{C min}^{-1}$ to $2.04^\circ\text{C min}^{-1}$) suggest that subsystem power draw significantly modulates the ramp slope, with the

gyro test showing the steepest rise due to the sustained ADCS peripheral load.

The 21 stuck thermistors in 8 of 9 sessions indicate an unresolved hardware or firmware issue with the expansion thermistor interface on the March 12 hardware configuration. The fact that CameraTest_1 (22 March) shows only one stuck channel suggests the fault was partially corrected between campaigns. These channels are flagged deterministically and excluded from statistical analyses.

The voltage rail anomalies (Table ??) appear correlated with ADCS peripheral load rather than representing true power faults. This contextual dependency is an important consideration for ML classifier design: a model trained on Baseline_1 (low z-score counts) may over-flag normal ADCS-on behaviour in subsequent sessions. The synthetic injection dataset mitigates this by generating fault examples from the relevant baseline sessions.

B. Limitations

Several limitations constrain the scope of the current study:

- 1) **Session duration:** all sessions are 3–10 minutes, limiting the ability to characterise slow-drift anomalies that manifest over longer timescales.
- 2) **Missing ThermalTest_1:** both files for this session are empty (1 line), representing a logging capture failure. Re-running the session is recommended.
- 3) **Debug timestamp resolution:** second-level timestamps in Debug files preclude packet-level timing analysis and result in $\sim 50\%$ row loss after deduplication for some sessions.
- 4) **Injector realism:** synthetic injections are applied as mathematical transformations and do not account for secondary physical effects (e.g. a power sag would also affect downstream temperature trajectories). Higher-fidelity injection would require a coupled physics model.

C. Future Work

Immediate next steps include: (i) training and evaluating LSTM and Isolation Forest classifiers on the labelled injection dataset; (ii) extending session durations to 30–60 minutes to capture thermal equilibrium behaviour; (iii) implementing a higher-resolution Debug file parser that disambiguates sub-second packets using the firmware SECOND counter field; and (iv) adding a detection module for magnetorquer / reaction wheel coupling anomalies using the wheel–gyro rolling correlation already computed by the pipeline.

IX. CONCLUSION

This paper presented a complete telemetry analysis pipeline for the KISPE Satellite Learning Laboratory, applied to nine subsystem test campaigns. The pipeline ingests both Analysis and Debug SCOTTI archive formats, applies four complementary anomaly detectors, and generates rich diagnostic visualisations — all fully automated. Key empirical findings include a universal OBDH thermal ramp of $1.40\text{ }^{\circ}\text{C min}^{-1}$ on average,

a systemic 21-channel thermistor interface fault, experiment-dependent voltage transients correlated with ADCS peripheral loading, and two frozen ADCS sensor registers. To support data-driven fault detection research, a 16-class synthetic fault injector produces 639 labelled telemetry artefacts across all campaigns from real hardware baseline data. The pipeline and dataset are made openly available.